# Nabbihni | Product Expiry Tracker

**Use Cases and UML Lab**
Effat University - CS3151 Software Engineering - Section 1

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## System scope

A bilingual mobile app that stores manually entered product expiry dates, organizes products by category, displays expiry status, and schedules local reminders. The prototype uses one local product collection on the device. Login, cloud sharing, barcode scanning, payments, and advertisements are outside this lab scope.

## Public GitHub repository

github.com/DanaEshmawi19/Software-Engineereing

Previous labs and project documents are available in our public repository.

## Introduction

Our project helps users track expiry dates for food, medicine, supplements, makeup, skincare, hair products and household items. Users add products, organize categories and receive reminders in Arabic or English.

# D1 | Actors and use case priorities

| Actor | Description / persona mapping | Goals and constraints |
|---|---|---|
| App User | Layan: student; Abeer: household shopper; Rawan: beauty-product user. All share the same role. | Manage product records, organize products, check expiry, and use Arabic or English. Dates must be entered accurately. |
| Phone Notification Service | External phone service used for local reminders. | Report permission and schedule/cancel reminders. Delivery timing remains subject to phone settings. |

| ID | Use case | Actor | Goal / outcome | Priority |
|---|---|---|---|---|
| UC-1 | Add Product | App User | Save a product and optional reminder | M |
| UC-2 | Edit Product | App User | Correct product details and replace its reminder | M |
| UC-3 | Delete Product | App User | Remove a product and cancel its reminder | M |
| UC-4 | Browse Products | App User | View products, categories and expiry status | M |
| UC-5 | View Upcoming Expiry | App User | See products expiring in the next seven days | M |
| UC-6 | Search / Filter Products | App User | Find products by name or category | S |
| UC-7 | Change Language | App User | Use Arabic or English | M |
| UC-8 | Configure Reminder | App User | Enable, disable or change a product reminder | M |

M = Must have; S = Should have. Phone Notification Service supports reminder scheduling and cancellation.

Won't have in the prototype: barcode scanning, shared cloud accounts, subscriptions and retailer integration.

# D1 | Requirements and traceability

| Req | Requirement / source | Use cases |
|---|---|---|
| R1 | Save product name, category and valid expiry date. Source: earlier Lab 3 REQ-6; Layan scenario. | UC-1 |
| R2 | Edit product details and persist changes. Source: feasibility product records and acceptance checks. | UC-2 |
| R3 | Delete product and cancel its reminder. Source: Rawan scenario; feasibility acceptance checks. | UC-3 |
| R4 | Organize products by category, including separate makeup and skincare. Source: earlier Lab 3 REQ-3 / REQ-5; Abeer and Rawan. | UC-1, UC-2, UC-4, UC-6 |
| R5 | Show products expiring within seven days in nearest-expiry order and identify expired items. Source: earlier Lab 3 REQ-4; feasibility expiry logic. | UC-4, UC-5 |
| R6 | Search names and filter by category. Source: Rawan persona feature; feasibility functions. | UC-6 |
| R7 | Switch core screens between Arabic and English. Source: earlier Lab 3 REQ-1; Layan persona. | UC-7 |
| R8 | Schedule reminders; show permission status; update or cancel reminders when products change. Source: earlier Lab 3 REQ-2; feasibility notifications. | UC-1, UC-2, UC-3, UC-8 |

## Non-functional requirements

N1: After successful saving, records and language preference remain after restarting the app. N2: All eight core use cases have Arabic and English text; Arabic screens use right-to-left layout. N3: Lists show results within two seconds for a collection of up to 100 products on the chosen test phone. N4: Saving works without internet. N5: A failed or denied reminder request is shown explicitly; the app must not claim it was scheduled.

# D2 | Brief use case specifications

| Use case | Preconditions | Success result / main flow |
|---|---|---|
| UC-1 Add Product | App open; categories available. | Enter details; validate; save; optionally configure reminder. Product appears in its category. |
| UC-2 Edit Product | A saved product exists. | Select product; change fields; save; cancel old reminder; optionally schedule a new one. |
| UC-3 Delete Product | A saved product exists. | Select product; confirm deletion; cancel reminder; remove record. |
| UC-4 Browse Products | App open. | Open collection; calculate status; display products by category. Empty collection shows a message. |
| UC-5 View Upcoming Expiry | App open. | Calculate remaining days; select dates from today to +7 days; sort nearest first. |
| UC-6 Search / Filter Products | App open. | Enter name or select category; show matching products or no-results message. |
| UC-7 Change Language | App open. | Select Arabic or English; save preference; update text and layout. |
| UC-8 Configure Reminder | A saved product exists. | Choose enabled state and lead time; validate schedule time; check permission; schedule or cancel. |

Common failure guarantee: invalid input does not create a product or overwrite an existing one. Cancellation before saving leaves the record unchanged. Persistence errors are reported to the user. Reminder failures do not hide the saved product.

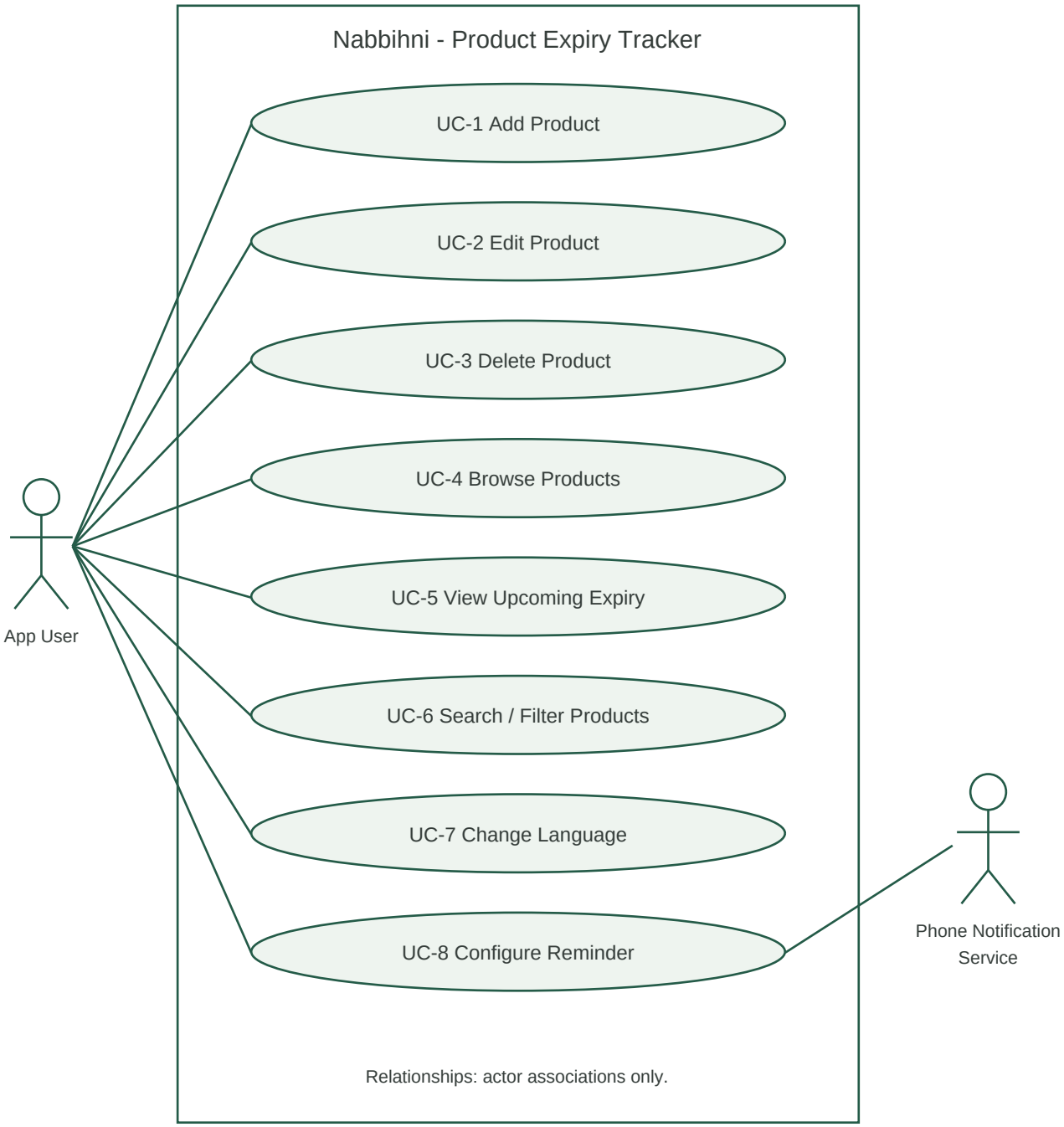
# D2 | UC-1 Add Product

| Field | Specification |
|---|---|
| Primary actor / trigger | App User selects Add Product. |
| Stakeholders | User wants accurate expiry tracking; Notification Service needs a valid future request and permission. |
| Preconditions | App is open. Supported categories are available. No login or internet connection is required. |
| Success guarantee | A new Product belongs to the ProductCollection and one Category. If a reminder was selected and accepted, its schedule and status are stored. |
| Minimum guarantee | No incomplete product is saved. A rejected reminder is never shown as active. |
| Main success scenario | 1. User enters name, category, expiry date and selects a reminder lead time.<br>2. App validates required fields and date.<br>3. App saves the product locally and receives its saved identifier.<br>4. App calculates a future reminder time.<br>5. App checks notification permission; permission is granted.<br>6. App asks Notification Service to schedule the reminder; service accepts.<br>7. App records reminder status and displays the saved product and confirmation. |
| Extensions | 2a. Missing name, unknown category or invalid date: show field error; return to step 1.<br>2b. Date is today or in the past: allow recording, label status correctly, and skip advance reminder.<br>3a. Saving fails: show error; create no reminder; user retries or cancels.<br>4a. Reminder disabled: skip steps 4-6; save without an active reminder.<br>4b. Calculated time has passed: skip scheduling; show reminder unavailable and current expiry status.<br>5a. Permission denied: retain product; show reminder inactive and guidance to phone settings.<br>6a. Service rejects schedule: retain product; mark reminder failed; offer retry through UC-8. |
| Special requirements | N1, N2, N4 and N5 apply. Required fields: nonblank name, supported category, valid calendar date. |
| Constraints | Expiry dates are entered manually. Only future reminders are scheduled. |
| Open issues | Supported phone platform and reminder time options. |

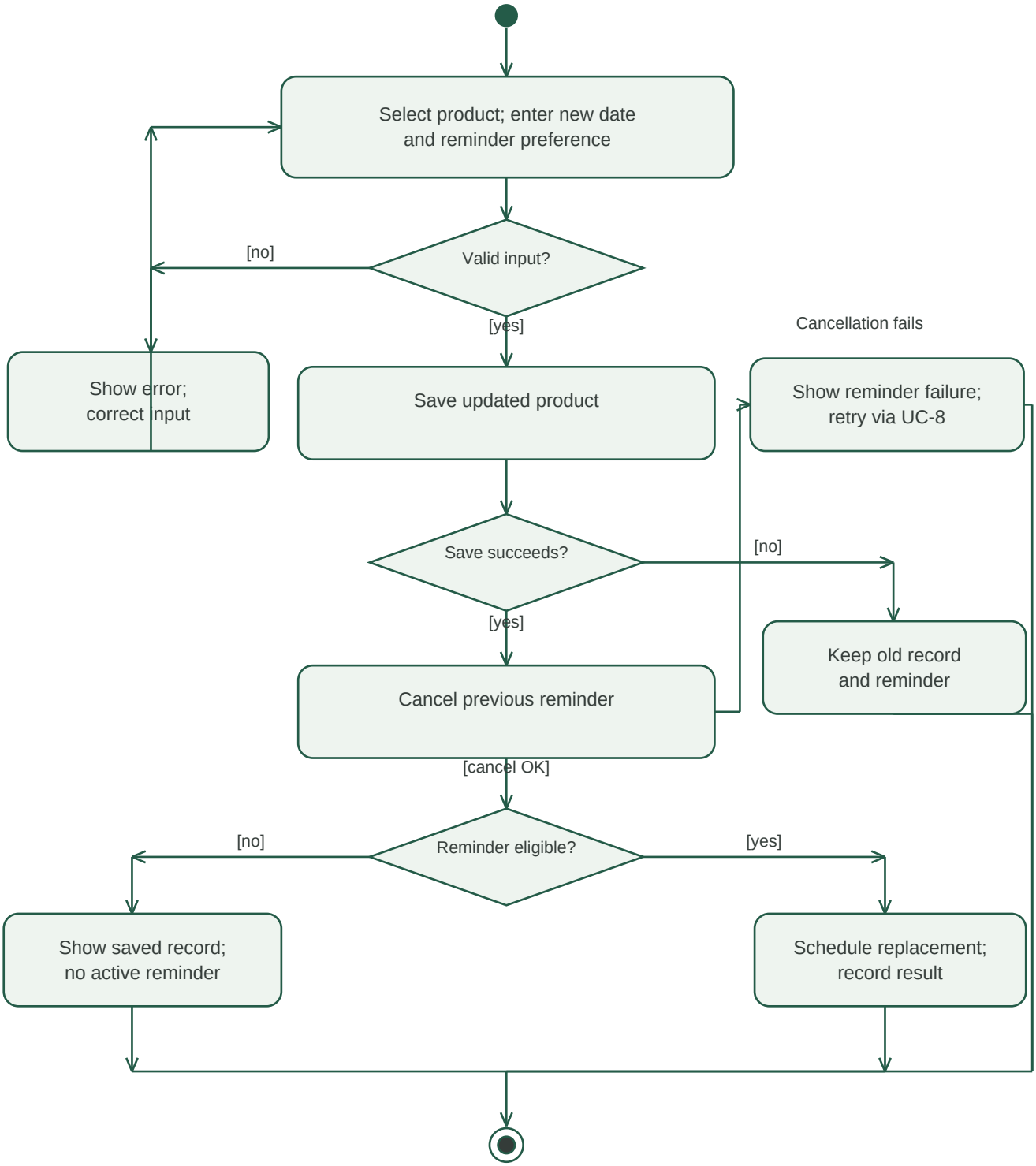
# D2 | UC-2 Edit Product

| Field | Specification |
|---|---|
| Primary actor / trigger | App User selects an existing product to edit its expiry date. |
| Stakeholders | User needs corrected information; Notification Service must not retain an outdated reminder. |
| Preconditions | Product exists in the local collection. Its current details and reminder state can be retrieved. |
| Success guarantee | Updated fields are stored; status is recalculated; the previous reminder is cancelled and a replacement is accepted when eligible. |
| Minimum guarantee | If validation or saving fails, the old record and reminder remain. If notification operations fail after saving, the saved record remains with a visible failure status. |
| Main success scenario | 1. User opens a product and changes its expiry date and reminder preference.<br>2. App validates the changed fields.<br>3. App saves updated product details.<br>4. App cancels the previous reminder, if any.<br>5. App checks whether the selected reminder is enabled, in the future and permitted.<br>6. Notification Service accepts a replacement reminder.<br>7. App stores reminder status and displays updated product and expiry status. |
| Extensions | 2a. Invalid fields: show error and return to step 1.<br>3a. Saving fails: preserve original record and original reminder; report failure.<br>4a. Cancellation fails: do not schedule a replacement; mark notification update failed and offer retry. Warn that an old alert may still appear.<br>5a. Reminder disabled, expired/today date, past reminder time or permission denied: keep updated product, leave no active reminder after successful cancellation, and explain why.<br>6a. Scheduling fails: keep updated product; mark reminder failed; offer retry through UC-8.<br>1a. User cancels before saving: end with no change. |
| Special requirements | N1, N2, N4 and N5 apply. Reuse product identity when replacing a reminder to avoid duplicate alerts. |
| Constraints | Save updated details before changing the reminder. |
| Open issues | Notification cancellation behavior on the supported platform. |

# D3 | Use case diagram

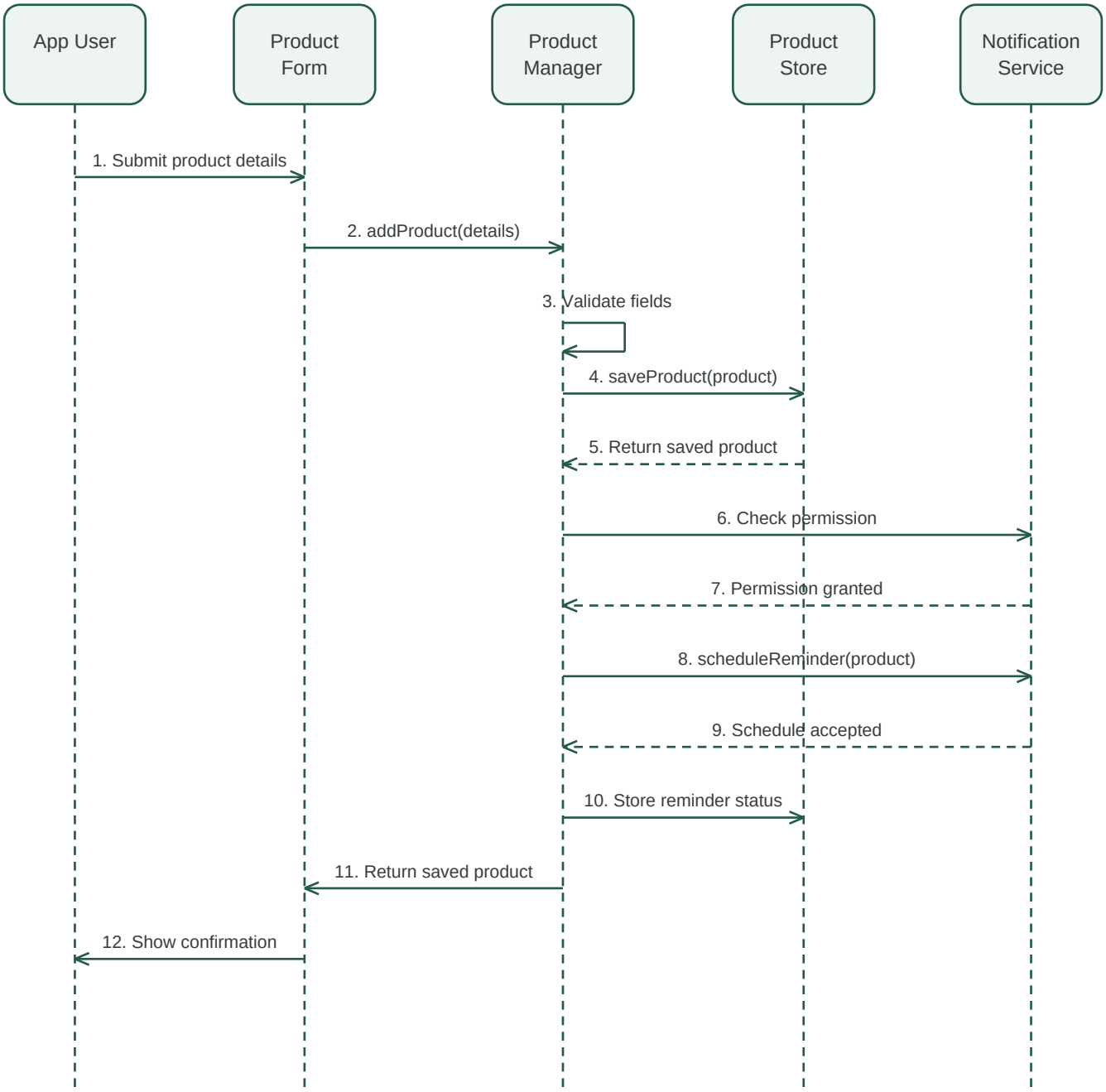


# D4 | Activity diagram: Edit Product




Eligible = enabled AND future schedule time AND notification permission granted. A scheduling or cancellation failure follows the failure branch in UC-2: retain the saved product, show a visible failure and offer retry.

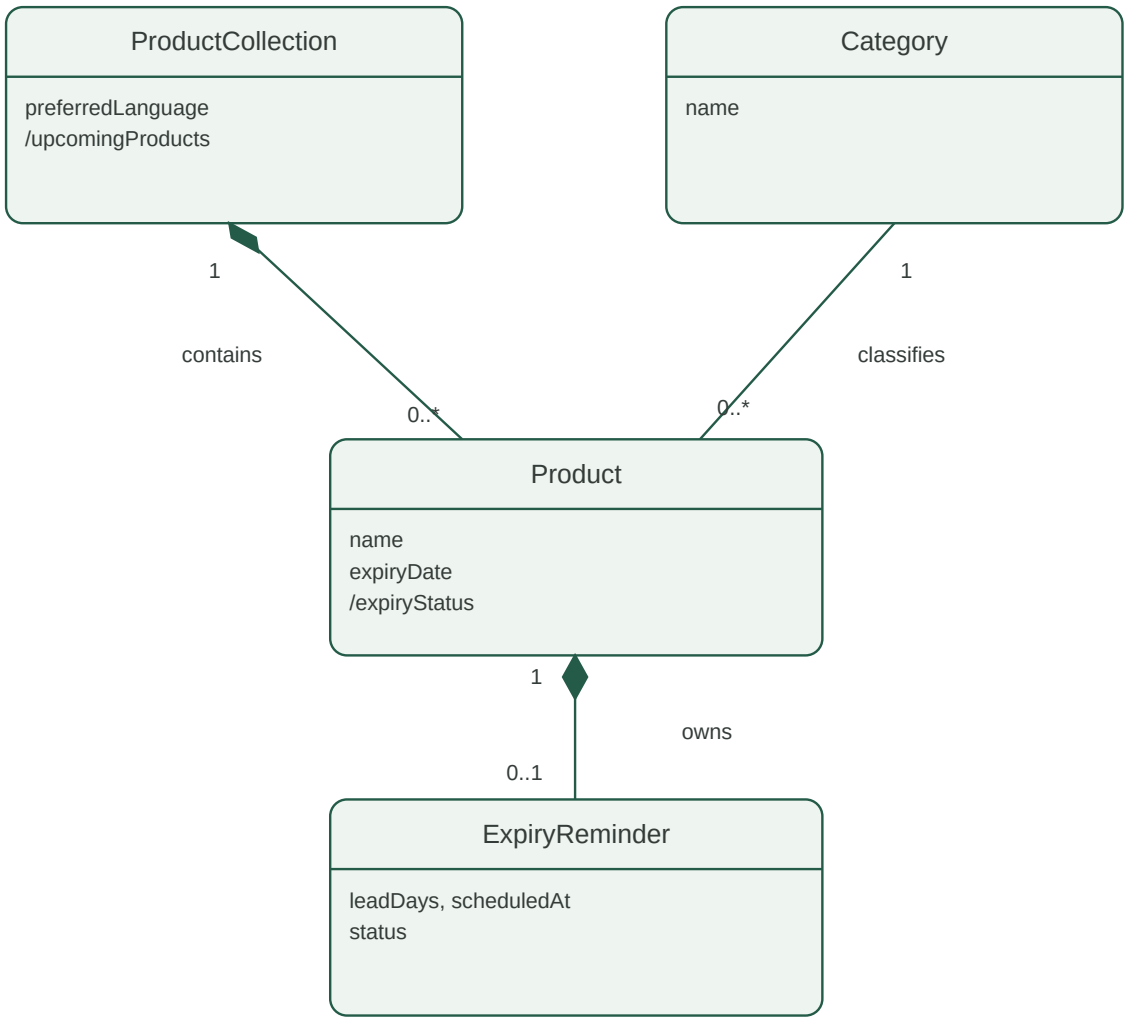
# D5 | Sequence diagram: Add Product




Main success scenario: valid future date, future reminder time, granted permission and accepted schedule. Product Manager calculates the reminder time during validation.

N4 note: Product Store is local, so product saving requires no internet. N5 note: confirmation of an active reminder occurs only after the phone service accepts it.

# D6 | Domain model / class diagram




| Concept | Meaning |
|---|---|
| ProductCollection | The device-local collection and preferred language. Owns zero or more products. |
| Product | A tracked item with a name and entered expiry date. /expiryStatus is derived from the current date. |
| Category | A classification such as food, medicine, supplements, makeup, skincare, hair or household. Each product has exactly one category. |
| ExpiryReminder | Optional reminder configuration and outcome owned by one product. Includes selected lead days, schedule time and status. |

Filled diamonds indicate composition: deleting a product removes its reminder; the collection owns its products. A category may classify zero or more products and remains available when a product is deleted. Multiplicities: collection 1 to products 0..*; category 1 to products 0..*; product 1 to reminder 0..1.

# D7 | Consistency and review

| Check | Evidence / result |
|---|---|
| Actor coverage | All three personas map to App User; external Notification Service is represented. |
| Use case coverage | UC-1 through UC-8 each have App User association and appear in D3. |
| Traceability | R1-R8 map to use cases in D1; persona and feasibility sources are named. |
| Sequence consistency | D5 follows UC-1: validate, save, check permission, schedule, record status, confirm. |
| Activity consistency | D4 follows UC-2 input validation, save failure, reminder replacement and eligibility branches. Cancellation/scheduling failures are clarified in its note. |
| Domain consistency | Collection, Product and Category appear in product management narratives; ExpiryReminder appears in UC-1, UC-2 and UC-8. |
| Constraints consistency | Persistence and offline saving are noted in D5; reminder eligibility guards appear in D4. Bilingual text applies to all narratives. |
| Notation and scope | Actors remain outside boundary; composition and multiplicities are stated; no unjustified include/extend relations. |

## Peer-review sheet

Reviewer team: Group 4, Section 1
Reviewer names: Ali Ahmed, Mohammed Guail, Abdulrahman Al Shalhoub

✓ Pass [ ] Revise - Actors cover personas and external systems

✓ Pass [ ] Revise - Use case names are clear user goals

✓ Pass [ ] Revise - Include / extend relationships are appropriate or unnecessary

✓ Pass [ ] Revise - Activity and sequence diagrams match the narratives

✓ Pass [ ] Revise - Domain concepts and multiplicities make sense

✓ Pass [ ] Revise - Preconditions and success guarantees are clear

✓ Pass [ ] Revise - All artifacts are consistent

✓ Pass [ ] Revise - Layout and labels are clear